

Life Processes (Set B) — MCQ Worksheet

Science • Chapter 5 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. Autotrophic organisms include:

- (A) Green plants and some bacteria
- (B) Mushroom
- (C) Tapeworm
- (D) Lion

Q2. Chlorophyll is needed for:

- (A) Absorbing sunlight for photosynthesis
- (B) Cell division
- (C) Cell wall formation
- (D) Storage of food

Q3. Stomatal opening and closing is controlled by:

- (A) Guard cells (changes in turgidity)
- (B) Mesophyll
- (C) Phloem
- (D) Xylem

Q4. Salivary amylase digests:

- (A) Starch into sugars
- (B) Proteins
- (C) Fats
- (D) Sugars

Q5. Pepsin works best in:

- (A) Acidic medium of stomach
- (B) Basic medium
- (C) Neutral
- (D) Hot water

Q6. Trypsin is secreted by:

- (A) Pancreas
- (B) Liver
- (C) Stomach
- (D) Salivary glands

Q7. Villi increase the:

- (A) Surface area for absorption in intestine
- (B) Length of stomach
- (C) Acidity of stomach
- (D) Speed of food

Q8. Anaerobic respiration in yeast produces:

- (A) Ethanol and CO₂
- (B) Lactic acid
- (C) Acetic acid
- (D) Pure water

Q9. Functional unit of human kidney is:

- (A) Nephron
- (B) Neuron
- (C) Alveolus
- (D) Hepatocyte

Q10. Glomerulus is in:

- (A) Bowman's capsule of nephron
- (B) Stomach
- (C) Lungs
- (D) Heart

Q11. RBCs:

- (A) Contain haemoglobin and carry oxygen
- (B) Contain antibodies
- (C) Form clots
- (D) Are nucleated in adults

Q12. Lymph differs from blood in not having:

- (A) Red blood cells
- (B) Plasma
- (C) White blood cells
- (D) Water

Q13. In humans, double circulation means blood passes through heart:

- (A) Twice in one cycle
- (B) Once
- (C) Three times
- (D) Four times

Q14. Xylem is responsible for transport of:

- (A) Water and minerals upwards
- (B) Food
- (C) Hormones
- (D) Oxygen

Q15. Phloem transport of food (translocation) requires:

- (A) Energy (active process)
- (C) Sun only

- (B) No energy
- (D) Air

Q16. Why are alveoli structured as numerous small sacs in lungs?

- (A) To increase surface area for gas exchange
- (B) To trap dust
- (C) To produce mucus
- (D) To filter blood

Q17. In photosynthesis, light reactions occur in:

- (A) Thylakoid membranes of chloroplast
- (B) Stroma
- (C) Cytoplasm
- (D) Mitochondria

Q18. The dark reactions of photosynthesis fix:

- (A) CO₂ into carbohydrates
- (B) Oxygen
- (C) Hydrogen
- (D) Nitrogen

Q19. Bile aids in digestion of:

- (A) Fats (emulsification)
- (B) Proteins
- (C) Carbohydrates
- (D) Vitamins only

Q20. Why is plant transport slow compared to animal blood circulation?

- (A) Plants lack a pumping organ; transport relies on transpiration pull and osmosis
- (B) Plants are dead
- (C) Plants have no need for fast transport
- (D) Cell walls block flow

Q21. Excretion in plants involves:

- (A) Storing waste in old tissues, releasing gases through stomata, leaf fall
- (B) Direct urine excretion
- (C) Faeces
- (D) Sweat

Q22. In humans, oxygenated and deoxygenated blood do not mix because of:

- (A) Septum dividing the heart into left and right halves
- (B) Different colour valves
- (C) Air gaps
- (D) Lungs only

Q23. Selective reabsorption in kidney is for:

- (A) Glucose, amino acids, water
- (B) Urea only
- (C) Excess salts only
- (D) Nothing

Q24. The site of cellular respiration is:

- (A) Mitochondrion (aerobic stage)
- (B) Nucleus
- (C) Lysosome
- (D) Ribosome

Q25. Why is haemoglobin necessary?

- (A) Oxygen has very low solubility in plasma; haemoglobin carries most of it
- (B) It gives colour
- (C) It clots
- (D) It fights bacteria